
PHYSICS

9702/36

Paper 3 Advanced Practical Skills 2

October/November 2018

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

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This document consists of **8** printed pages.



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PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

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Question	Answer	Marks
1(a)	Value of L_0 with unit. Value on answer line in the range 1.50–2.50 cm.	1
1(b)	Value of θ with unit, to the nearest degree and $\theta \leq 90^\circ$.	1
1(c)	Six sets of readings of θ and L with the correct trend and without help from the Supervisor scores 5 marks, five sets scores 4 marks, etc.	5
	Range: $\theta_{\max} \geq 30^\circ$ and $\theta_{\min} \leq 15^\circ$.	1
	Column headings: Each column heading must contain a quantity and a unit where appropriate. The presentation of quantity and unit must conform to accepted scientific convention e.g. L/mm . There must be no unit for $(\sin \theta)(\cos \theta)$.	1
	Consistency: All values of L must be given to the nearest mm only.	1
	Significant figures: Number of significant figures for every value of $(\sin \theta)(\cos \theta)$ same as, or one greater than, the number of s.f. of θ as recorded in table.	1
	Calculation: Values of $(\sin \theta)(\cos \theta)$ calculated correctly.	1

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Question	Answer	Marks
1(d)(i)	Axes: Sensible scales must be used, no awkward scales (e.g. 3:10 or fractions). Scales must be chosen so that the plotted points occupy at least half the graph grid in both x and y directions. Scales must be labelled with the quantity that is being plotted. Scale markings should be no more than three large squares apart.	1
	Plotting of points: All observations in the table must be plotted on the grid. Diameter of plotted points must be $\leq$ half a small square (no “blobs”). Points must be plotted to an accuracy of half a small square.	1
	Quality: General trend of points on graph must be positive. All points in the table (at least 5) must be plotted on the grid. It must be possible to draw a straight line that is within ± 0.05 on the x-axis of all plotted points.	1
1(d)(ii)	Line of best fit: Judge by balance of all points on the grid about the candidate's line (at least 5 points). There must be an even distribution of points either side of the line along the full length. Allow one anomalous point only if clearly indicated by the candidate (i.e. circled or labelled). Line must not be kinked or thicker than half a small square.	1
1(d)(iii)	Gradient: The hypotenuse of the triangle used must be greater than half the length of the drawn line. The method of calculation must be correct. Do not allow $\Delta x / \Delta y$. Both read-offs must be accurate to half a small square in both the x and y directions. Sign of gradient on answer line must match graph.	1
	y-intercept: Correct read-off from a point on the line substituted into $y = mx + c$. Read-off must be accurate to half a small square in both x and y directions. or Intercept read directly from the graph with read-off at $x = 0$, accurate to half a small square.	1

Question	Answer	Marks
1(e)	Value of a equal to candidate's gradient and given to two or more significant figures. and Value of b equal to candidate's intercept. The values must not be fractions.	1
	Unit for a correct and unit for b correct (e.g. m, cm, mm).	1

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Question	Answer	Marks
2(a)	Value of h_0 with unit and to the nearest mm.	1
2(b)(i)	Value of h less than h_0 .	1
2(b)(ii)	Correct calculation of y , with unit.	1
2(c)	Percentage uncertainty based on an absolute uncertainty in y of 3–6 mm. If repeated readings have been taken, then the uncertainty can be half the range (but not zero) if the working is clearly shown. Correct method of calculation to obtain percentage uncertainty.	1
2(d)	Value for T on answer line in range 0.2–0.9 s, with unit.	1
	Evidence of repeat readings of time. There must be at least two measurements of nT where $n \geq 5$.	1
2(e)	Second value of h .	1
	Second value of T .	1
	Quality: T greater for greater y .	1
2(f)(i)	Two values of c calculated correctly.	1
2(f)(ii)	Valid comment consistent with the calculated values of c , testing against a criterion specified by the candidate.	1
2(g)	Correct calculation of g with consistent unit.	1

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Question	Answer	Marks
2(h)(i)	A Two readings are not enough to draw a (valid) conclusion (not “not enough for accurate results”, “few readings”). B Rule may not be vertical (when measuring heights). C Difficult to measure h_0 because blade not horizontal/blade is bent/h_0 varies along blade. D Difficult to measure h because mass is in the way/parallax when measuring h/difficult to judge position of the centre of the mass. E Difficult to determine T with a reason e.g. difficult to judge start/end/completion of oscillations or difficult to count oscillations. <i>1 mark for each point up to a maximum of 4.</i>	4
2(h)(ii)	A Take many readings <u>and</u> plot a graph or take more values of c <u>and</u> compare (not “repeat readings” on its own). B Method to check rule vertical, e.g. use set square <u>on floor</u>/use plumb line/use spirit level. C Mark position for centre of the mass on blade then measure h_0 at that position. D Improved method of measuring h, e.g. (clamp) rule and use set square <u>as pointer</u>/hang mass from thread/use a mass that is narrower than the blade/measure blade height at both sides of mass and average. E Video/film/record with timer in view (or use frame-by-frame)/use motion sensor above (or below) the blade or use larger masses to give longer T/to make counting easier. <i>1 mark for each point up to a maximum of 4.</i>	4